Searching algorithms: linear and binary search

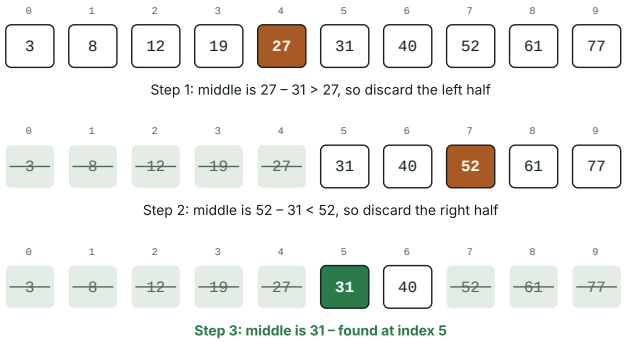
D12 FACT FILE

Specification: students should be able to

- Create and evaluate algorithms, including those for basic searching
- Simple searching techniques such as linear and binary searching

CCEA fact file: Programming Constructs 2

Key words: search, target, linear search, binary search, midpoint, sorted, efficiency, found flag



Binary search for 31 in a sorted list, halving the search each step

Linear search

A linear search checks each item in turn from the start until the target is found or every item has been checked. It works on any list, sorted or not, but on average it must look at half the list and in the worst case all of it, so it is slow for large data sets.

Binary search

A binary search only works on sorted data. It compares the target with the middle item; if they match the search ends, otherwise it discards the half that cannot contain the target and repeats on the other half. Because the list halves each time, it finds items in very few steps even in large lists.

T Key terms from the mark schemes

Linear search
Checks each item in turn until the target is found or the list ends.
Credited: U4 2019 Q12, 2024 Q7

Binary search
Repeatedly checks the middle of a sorted list and discards half; needs sorted data.
Credited: U4 2019 Q12, 2024 Q7

C C# example

Linear search

```
static int LinearSearch(int[] items, int target)
{
    for (int i = 0; i < items.Length; i++)
    {
        if (items[i] == target)
        {
            return i;
        }
    }
    return -1;
}

int[] data = { 12, 7, 31, 4 };
Console.WriteLine(LinearSearch(data, 31));   // 2
```

Exam focus

Efficiency comparison (2022 Q1(f)), definitions and advantages of binary search (2019 Q12), describing both and the sorted-data condition (2024 Q7(e)), choosing the efficient method for sorted data (2023 Q9(c)(iii)).

Unit 4 2024 Q7	[35]	Arrays, pseudocode, FOR/WHILE totals, minimum, bubble sort, linear and binary search
Unit 4 2023 Q9	[18]	Arrays: update, reset to zero, find maximum; bubble sort steps; searching sorted data
Unit 4 2022 Q1(d)–(g)	[4]	Abstraction, decomposition, search efficiency, ASCII
Unit 4 2019 Q12	[8]	Linear vs binary search; showing sort passes

? Check yourself

1. Describe how a linear search finds an item. [2]
2. Describe how a binary search finds an item. [3]
3. State the condition that must be met before a binary search can be used. [1]

Answers: 1. Start at the first item and check each item in turn (1) / until the target is found or the end of the list is reached (1) 2. Look at the middle item; if it is the target, it is found (1) / if the target is smaller discard the upper half, if larger discard the lower half (1) / repeat on the remaining half until found or no items are left (1) 3. The data must be sorted (in order) first (1)